

Quadratic Approximation Horizon for Finite Riemann–Siegel Reconstruction of the Critical Line

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June 1, 2026

Abstract

We quantify how far a finite arithmetic of N integers approximates the Hardy function $Z(t)$ and, under explicit stability assumptions, reconstructs its critical-line zeros, distinguishing three levels of claim. First, and rigorously, the functional-equation-folded approximant built from the Riemann–Siegel main sum agrees with the Hardy function $Z(t)$ up to the remainder $O(t^{-1/4})$ for all $t \leq 2\pi N^2$: the budget N therefore buys a quadratic approximation horizon $\sim 2\pi N^2$, and we show this horizon is sharp in mean square. Second, and conditionally, under simplicity of the critical-line zeros and a local stability condition on Z between them, the same approximant faithfully reconstructs every such zero up to that horizon; we make the two conditions explicit and exhibit, numerically, the sparse close pairs where the second condition fails. Third, and heuristically, the literal one-sided Dirichlet truncation reconstructs only to a linear scale: a zero-counting argument bounds its reach by $2\pi e N$, and experiments place the failure near $2\pi N$. Direct reconstruction experiments confirm the picture. The results concern reproduction of zeros already known to lie on the line; they say nothing about the real parts of the zeros, and are independent of the Riemann Hypothesis.

Keywords: Riemann zeta function; critical line; approximate functional equation; Riemann–Siegel formula; Dirichlet polynomial; Gauss sums.

1 Introduction

The practical algorithms for locating zeros of the Riemann zeta function on the critical line, above all the Riemann–Siegel formula [1–3], use remarkably few arithmetic terms relative to the height they reach: to evaluate near height t one sums about $\sqrt{t/2\pi}$ terms. Read backwards, a budget of N integers corresponds to a height scale $t \sim 2\pi N^2$. This note isolates that correspondence, states precisely what it does and does not give, and contrasts it with the literal one-sided Dirichlet truncation.

Two approximants. On the critical line write $s = \frac{1}{2} + it$. The literal partial sum of the Dirichlet series is

$$\zeta_N(s) = \sum_{n=1}^N n^{-s}. \quad (1)$$

Let $\theta(t)$ be the Riemann–Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ the (real-valued) Hardy function, whose real zeros are exactly the critical-line zeros of ζ . With $\nu(t) = \lfloor \sqrt{t/2\pi} \rfloor$, define the *budget- N folded approximant*

$$Z_N(t) = 2 \sum_{n=1}^{\min(\nu(t), N)} \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}}. \quad (2)$$

For $\nu(t) \leq N$ this is exactly the Riemann–Siegel main sum; the cap at N encodes the finite budget.

Results, in three levels. We are deliberately careful to separate what is proved from what is conditional or heuristic.

Level 1 (rigorous): a quadratic approximation horizon. The folded approximant satisfies $Z_N(t) = Z(t) + O(t^{-1/4})$ uniformly on $t \leq 2\pi N^2$ (Theorem 3), and this horizon is sharp in mean square: beyond $2\pi N^2$ the mean-square error grows without bound (Proposition 4).

Level 2 (conditional): faithful zero reconstruction up to the horizon. Under simplicity of the critical-line zeros and a local stability and separation condition on Z (Conditions (A) and (B)), the folded approximant reconstructs every critical-line zero up to $2\pi N^2$ (Theorem 5). The second condition can fail at unusually close (Lehmer) pairs, and we locate one such pair numerically.

Level 3 (heuristic and numerical): a linear barrier for the literal truncation. The literal sum (1) is an exponential polynomial of bandwidth $\log N$; under a natural detection assumption a zero-counting argument bounds its reach by $2\pi e N(1 + o(1))$ (Heuristic 1), and experiments locate the failure near $2\pi N$.

Relation to known results. The analytic ingredients are classical: the approximate functional equation [4, 5], the Riemann–Siegel main sum and its remainder [1, 2], the mean value theorem for Dirichlet polynomials [6], and the density of zeros of exponential polynomials [7]. The contribution here is organisational and quantitative: a clean budget-to-height law for the folded approximant, proved at the level of approximation and sharp in mean square; an explicit, conditional reconstruction statement with its hypotheses isolated and tested; and a contrasting linear barrier for the literal truncation. We make no claim to new analytic technique.

2 Framework

Throughout, $\gamma > 0$ denotes the ordinate of a zero of the Hardy function Z , equivalently a critical-line zero of ζ ; we make no assumption about any zeros off the critical line. The Hardy function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is real for real t , with $\theta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$; a real zero of Z is exactly a zero of ζ on the critical line [5, §4.17]. We use two counting functions: the classical Riemann–von Mangoldt count of all nontrivial zeros in the critical strip,

$$N_{\text{all}}(T) = \#\{\rho = \beta + i\gamma : 0 < \gamma \leq T\} = \frac{T}{2\pi} \log \frac{T}{2\pi e} + \frac{7}{8} + O(\log T), \quad (3)$$

and the count of critical-line zeros, $N_0(T) = \#\{\gamma \leq T : Z(\gamma) = 0\}$. Always $N_0(T) \leq N_{\text{all}}(T)$; a positive proportion of zeros lie on the line [8], so $N_0(T) \asymp N_{\text{all}}(T)$ unconditionally, and $N_0(T) = N_{\text{all}}(T)$ under the Riemann Hypothesis and on the verified range $T \leq 3 \cdot 10^{12}$ [9]. The reconstruction results below concern N_0 —the zeros actually on the line—and are in that sense independent of RH; the counting heuristic of Section 5 compares against $N_0(T)$, and attains its clean form when $N_0(T) \sim N_{\text{all}}(T)$.

We make “reconstruction” precise in a way that is robust to multiplicity. For a real approximant A (the folded case) a *detection* is a sign change of A or a local minimum of $|A|$ lying below a fixed fraction of the local average; for the complex literal sum it is a sufficiently deep local minimum of $|\zeta_N(\frac{1}{2} + it)|$. The use of local minima, rather than sign changes alone, means an even-multiplicity zero would still be detected.

Definition 1 (Faithful reconstruction). Fix a finite range $[T_0, T]$ and a tolerance $\delta > 0$ smaller than half the least gap between consecutive critical-line ordinates in $[T_0, T]$. An approximant A *faithfully reconstructs the critical-line zeros on* $[T_0, T]$ if every ordinate $\gamma \in [T_0, T]$ has a

detection of A within δ of it, and no detection of A in $[T_0, T]$ lies farther than δ from every ordinate (no spurious detections). In the numerical matching of Section 6 we take δ to be half the local mean spacing; the theoretical tolerance here is a fixed number for the given range.

Definition 1 requires reconstruction of *every* critical-line ordinate in the range; the Level-2 theorem below supplies exactly the hypotheses under which this holds, so there is no tension between the definition and the result.

3 Level 1: a quadratic approximation horizon

Lemma 2 (Riemann–Siegel main sum). *There is an absolute constant C_0 such that, with $\nu(t) = \lfloor \sqrt{t/2\pi} \rfloor$,*

$$Z(t) = 2 \sum_{n \leq \nu(t)} \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} + R(t), \quad |R(t)| \leq C_0 t^{-1/4},$$

uniformly for $t \geq 2$.

Proof. This is the Riemann–Siegel formula [1, 3], with explicit remainder bounds in (author?) [2]; it is the symmetric approximate functional equation of (author?) [4] on $\sigma = \frac{1}{2}$ with $x = y = \sqrt{t/2\pi}$, rotated by $e^{i\theta(t)}$. $\square$

Theorem 3 (Approximation horizon). *With C_0 as in Lemma 2, the folded approximant coincides with the full Riemann–Siegel main sum for all $t < 2\pi(N+1)^2$, and in particular*

$$|Z_N(t) - Z(t)| \leq C_0 t^{-1/4} \quad \text{for all } 2 \leq t \leq 2\pi N^2.$$

Thus for any fixed warm-up height $T_0 \geq 2$ the error is uniformly bounded by $C_0 T_0^{-1/4}$ on $[T_0, 2\pi N^2]$, and the approximation becomes uniformly accurate only on ranges whose lower endpoint also tends to infinity (the bound does not improve at the fixed left end $t = 2$, where the main sum merely stabilises). The scale $2\pi N^2$ is, to leading order, the height $2\pi(N+1)^2 = 2\pi N^2(1 + O(1/N))$ at which the budget cap first becomes active.

Proof. The cap in (2) is inactive exactly when $\nu(t) \leq N$, i.e. when $\sqrt{t/2\pi} < N+1$, i.e. for $t < 2\pi(N+1)^2$; there $\min(\nu(t), N) = \nu(t)$ and (2) is the full main sum, so Lemma 2 gives the bound. Since $2\pi N^2 < 2\pi(N+1)^2$, the displayed inequality holds throughout $[2, 2\pi N^2]$. $\square$

The horizon is sharp, in the mean-square sense, beyond $2\pi N^2$.

Proposition 4 (Mean-square sharpness). *Let $w \geq 0$ be smooth, supported in $[1, 2]$, with $\int w = 1$, and write $\langle f \rangle_T = \frac{1}{T} \int w(t/T) f(t) dt$. For fixed N , as $T \rightarrow \infty$,*

$$\langle |Z_N - Z|^2 \rangle_T = \log \frac{T}{2\pi N^2} (1 + o(1)) \xrightarrow{T \rightarrow \infty} \infty,$$

so no uniform approximation persists beyond the horizon, and the scale $2\pi N^2$ is sharp in mean square.

Proof. By Lemma 2, $Z - Z_N = B(t) + R(t)$ with $|R| \leq C_0 t^{-1/4}$ and the omitted block

$$B(t) = 2 \sum_{N < n \leq \nu(t)} \frac{\cos(\theta(t) - t \log n)}{\sqrt{n}} = 2 \operatorname{Re}[e^{i\theta(t)} S(t)], \quad S(t) = \sum_{N < n \leq \nu(t)} n^{-1/2 - it},$$

so $B^2 = 2|S|^2 + 2 \operatorname{Re}[e^{2i\theta} S^2]$.

Diagonal. The upper limit $\nu(t)$ is piecewise constant, so partition the support of w into the ranges $[2\pi m^2, 2\pi(m+1)^2)$ on which $\nu(t) \equiv m$. On each, S is a *fixed* Dirichlet polynomial

and the mean value theorem [6] gives $\int |S|^2 = (\text{length}) \sum_{N < n \leq m} n^{-1} + O(m)$, a relative error $O(1/\log(m/N))$. Since every contributing $m \geq \nu(T) = \lfloor \sqrt{T/2\pi} \rfloor \rightarrow \infty$ while N is fixed, this error is uniformly $o(1)$; summing the ranges, $\langle |S|^2 \rangle_T = \langle \sum_{N < n \leq \nu(t)} n^{-1} \rangle_T (1 + o(1))$.

Off-diagonal. In $\langle \text{Re}[e^{2i\theta} S^2] \rangle_T$ the phase $2\theta(t) - t \log(nn')$ has derivative $\log(t/2\pi nn')$. For pairs with $nn' \not\approx t/2\pi$ this is bounded away from 0 and the smoothness of w yields arbitrary savings by repeated integration by parts; the remaining pairs are those with stationary point $t^* = 2\pi nn'$ in $\text{supp } w(\cdot/T)$, a near-boundary set on which a standard stationary-phase estimate for Riemann–Siegel-type sums [5, Ch. 7] bounds the total by $o(\log T)$. Hence $\langle \text{Re}[e^{2i\theta} S^2] \rangle_T = o(\log T)$.

Combining, and using $\langle R^2 \rangle_T = O(T^{-1/2})$ with Cauchy–Schwarz for the cross term $\langle BR \rangle_T$, gives $\langle |Z_N - Z|^2 \rangle_T = 2 \langle \sum_{N < n \leq \nu(t)} n^{-1} \rangle_T (1 + o(1))$. Since $\sum_{N < n \leq \nu(t)} n^{-1} = \frac{1}{2} \log(t/2\pi N^2) + o(1)$, averaging over $[T, 2T]$ yields the stated asymptotic. $\square$

4 Level 2: conditional faithful reconstruction

Approximation in the sense of Theorem 3 becomes faithful reconstruction in the sense of Definition 1 only under control of the zeros and of the size of Z between them. We isolate the two hypotheses.

Condition (A). Every zero of Z in $(0, T]$ is simple (equivalently, every critical-line zero of ζ up to height T is simple). No assumption is made about possible zeros off the critical line.

Condition (B). Let C_0 be the constant of Lemma 2, let $\delta > 0$ be the tolerance of Definition 1, and let $\mathcal{Z} = \{\gamma : Z(\gamma) = 0\}$ be the set of critical-line ordinates. There is a warm-up height $T_0 \geq 2$ such that, on $[T_0, T]$: (i) *stability*— $|Z(t)| > C_0 t^{-1/4}$ whenever $\text{dist}(t, \mathcal{Z}) \geq \delta$; and (ii) *separation*—consecutive ordinates differ by more than 2δ .

Condition (B) is a local stability requirement: away from δ -neighbourhoods of the zeros, Z stays above the Riemann–Siegel remainder, and consecutive zeros are far enough apart to be told apart at resolution δ . Both parts can fail at unusually close (Lehmer) pairs [10], where Z remains small across the whole gap.

Theorem 5 (Conditional reconstruction). *Suppose Conditions (A) and (B) hold up to height T , with warm-up T_0 and tolerance δ . Then for every N with $2\pi N^2 \geq T$, the folded approximant Z_N faithfully reconstructs every critical-line ordinate in $[T_0, T]$.*

Proof. For $t \leq T \leq 2\pi N^2$ we have $\nu(t) \leq N$, so by Theorem 3 the approximant is the full main sum and $Z_N(t) = Z(t) + r(t)$ with $|r(t)| \leq C_0 t^{-1/4}$ on $[T_0, T]$. Since Condition (A) makes every critical-line zero simple, it suffices to use the sign-change detector for Z_N .

No spurious detection. If Z_N changes sign at some $t \in [T_0, T]$ then $Z_N(t) = 0$, so $|Z(t)| = |r(t)| \leq C_0 t^{-1/4}$, and stability (i) forces $\text{dist}(t, \mathcal{Z}) < \delta$. Thus every detection lies within δ of a critical-line ordinate.

Every ordinate detected. Let $\gamma \in [T_0, T]$ be an ordinate. The nearest other ordinate is, by separation (ii), more than 2δ away, so the points $\gamma \pm \delta$ satisfy $\text{dist}(\gamma \pm \delta, \mathcal{Z}) \geq \delta$ and $(\gamma - \delta, \gamma + \delta)$ contains no other ordinate. By stability (i), $|Z(\gamma \pm \delta)| > C_0 (\gamma \pm \delta)^{-1/4} \geq |r(\gamma \pm \delta)|$, so $Z_N(\gamma \pm \delta)$ has the same sign as $Z(\gamma \pm \delta)$; and these signs are opposite, since by Condition (A) the zero at γ is simple and there is no other zero in between. Hence Z_N changes sign in $(\gamma - \delta, \gamma + \delta)$, a detection within δ of γ .

These two facts are faithful reconstruction in the sense of Definition 1. $\square$

Remark 6 (Status of the conditions). Large-scale computations verify that all zeros up to $3 \cdot 10^{12}$ lie on the critical line [9]; the zeros encountered in such computations are simple in the numerical data, so Condition (A) holds there, although global simplicity remains unproven. Condition (B) holds for all numerically examined zeros except at rare close pairs; the experiments

of Section 6 exhibit one such pair, at ordinates 220.71 and 221.43 (gap 0.72, about 0.4 of the local mean spacing), which violates Condition (B): between the two zeros $|Z|$ does not rise above the Riemann–Siegel remainder, so the main sum—hence Z_N for every sufficiently large budget—cannot separate them. Removing Condition (B), i.e. controlling the small values of Z between close zeros unconditionally, is exactly the gap between Theorem 5 and an unconditional reconstruction statement.

5 Level 3: a linear barrier for the literal truncation

On the critical line the literal sum (1) is an exponential polynomial in t ,

$$\zeta_N(\tfrac{1}{2} + it) = \sum_{n=1}^N n^{-1/2} e^{-it \log n}, \quad (4)$$

with frequencies $\{\log n\}_{n \leq N} \subset [0, \log N]$. It is complex-valued on the real axis, so it has essentially no real zeros; the zeros that matter are those of the entire function (4) in a bounded strip about the real axis, and the critical-line zeros are detected through the dips of $|\zeta_N|$. This makes the following a heuristic upper bound on reach rather than a theorem about exact reconstruction.

Heuristic 1 (Counting bound for the literal truncation). *Assume that each deep local minimum of $|\zeta_N(\tfrac{1}{2} + it)|$ on the real axis corresponds to a zero of the entire function (4) within a bounded distance in a horizontal strip. The number of such zeros in the window $[0, T]$ is $\sim \frac{T}{2\pi} \log N$, so the number of detections of $|\zeta_N|$ in $[0, T]$ is at most $\frac{T}{2\pi} \log N (1 + o(1))$; faithful reconstruction of the $N_0(T)$ critical-line ordinates up to T then forces $\frac{T}{2\pi} \log N \gtrsim N_0(T)$. Since $N_0(T) \asymp \frac{T}{2\pi} \log T$ unconditionally, this gives a linear bound $T \leq C N$; and when $N_0(T) \sim N_{\text{all}}(T)$ (under RH or on the verified range) it sharpens to*

$$\log N \geq \log \frac{T}{2\pi e} (1 + o(1)), \quad \text{i.e.} \quad T \leq 2\pi e N (1 + o(1)).$$

Justification. The density of zeros of an exponential polynomial with frequencies spanning an interval of length $\log N$ is $\frac{1}{2\pi} \log N$ per unit length, by the mean-motion theory of almost periodic functions [7]; counting in $[0, T]$ gives the stated total. Under the detection assumption the number of detections is bounded by this count, and comparison with the critical-line count $N_0(T)$ (using $N_0(T) \sim N_{\text{all}}(T)$ from (3) for the sharp form) yields the inequality. The assumption itself—that real-axis dips of $|\zeta_N|$ are in bounded correspondence with strip zeros of (4)—is the unproved step, which is why this is recorded as a heuristic rather than a theorem. $\square$

Heuristic 1 concerns only the linear *exponent*, with the constant $2\pi e \approx 17.08$; it does not establish a sharp prefactor, nor does poor approximation by itself logically preclude accidental detections. The experiments below place the actual failure near $2\pi N \approx 6.28 N$, comfortably inside the counting bound. We therefore regard the linear scale as structural—fixed by the bandwidth $\log N$ of (4)—and the sharp constant 2π as an empirical observation rather than a theorem.

6 Numerical experiments

All zeta values use the arbitrary-precision routines of `mpmath` [11] (Z via `siegelz`, θ via `siegeltheta`); ordinates are obtained independently with `zetazero`; array work uses `NumPy` [12]. We test the three levels in turn.

N	horizon $2\pi N^2$	# zeros	matched	missed	spurious
3	56.5	12	11	1 (edge, 56.45)	0
4	100.5	29	29	0	0
5	157.1	56	56	0	0
6	226.2	94	92	2 (pair 220.71, 221.43)	0
7	307.9	143	141	2 (same pair)	0
8	402.1	203	201	2 (same pair)	0

Table 1: Direct reconstruction test (Definition 1) for the folded approximant up to its horizon. Reconstruction is faithful except at a single Lehmer-close pair, which the Riemann–Siegel main sum cannot separate for any budget; this is the failure of Condition (B).

Level 1: the approximation horizon. On a grid $t \in [2, 680]$ (step 0.2) we form, for each budget N , the capped approximant (2) and the literal sum (1), and record the height at which a smoothed approximation error first crosses a fixed threshold. The folded data follow a quadratic law in N (fitted exponent 1.97 against the predicted 2); the literal data follow a linear law (fitted slope 6.70). These small-budget fits confirm the *exponents* 2 and 1; the prefactor 2π is structural—it is the Riemann–Siegel term count $\nu(t) = \lfloor \sqrt{t/2\pi} \rfloor$, not a quantity inferred from the few data points. Figure 1 shows both against the reference curves $2\pi N$ and $2\pi N^2$. Consistent with Proposition 4, the folded mean-square error is negligible below $2\pi N^2$ and rises beyond it (for $N = 4$, with horizon 100.5, the windowed mean-square error grows from about 0.1 near $t = 150$ to order one by $t = 500$).

Level 2: direct reconstruction. We now test Definition 1 directly rather than through an error threshold. For each budget N we evaluate Z_N on a fine grid (step 0.01) up to the horizon $2\pi N^2$, extract its detections, match them to the true ordinates within half the local mean spacing, and count matched, missed and spurious detections. Table 1 reports the result. For $N = 4$ and $N = 5$ the reconstruction is perfect up to the horizon; for $N \geq 6$ the only failures are the two zeros of the single Lehmer pair at 220.71/221.43, which the Riemann–Siegel main sum cannot separate (Remark 6). There are no spurious detections. The lone miss at $N = 3$ sits at 56.45, at the very edge of the horizon 56.5, and is a boundary artefact. This is exactly the behaviour predicted by Theorem 5 together with the failure of Condition (B) at the close pair.

Level 3: the literal truncation. Figure 3 shows $|\zeta_5|$ and $|\zeta_{12}|$ tracking $|\zeta(\frac{1}{2} + it)|$ only while $t \lesssim 2\pi N$, then losing the zeros as the bandwidth $\log N$ falls below the local zero density—the mechanism of Heuristic 1. Direct matching of the dips of $|\zeta_N|$ confirms a linear failure scale near $2\pi N$, though the precise height is sensitive to the detection threshold, consistent with the heuristic status of the bound.

7 A finite analogue: Gauss sums and square-root normalisation

The economy underlying Level 1 is the symmetric approximate functional equation: the divergent conjugate tail of the one-sided sum is replaced by the reflected finite sum $\chi(\frac{1}{2} + it) \sum_{n \leq \sqrt{t/2\pi}} n^{-1/2+it}$, where $\chi(s) = \pi^{s-1/2} \Gamma(\frac{1-s}{2}) / \Gamma(\frac{s}{2})$ has $|\chi(\frac{1}{2} + it)| = 1$. The two finite halves, recombined into the real main sum (2), recover $Z(t)$ from $\sqrt{t/2\pi}$ terms rather than $\sim t/2\pi$; the square-root is the signature of the self-duality $s \mapsto 1 - s$.

It is worth recording, as a parallel rather than a consequence, that the same self-dual structure governs the simplest finite exponential sums. This statement is classical and is *not* derived

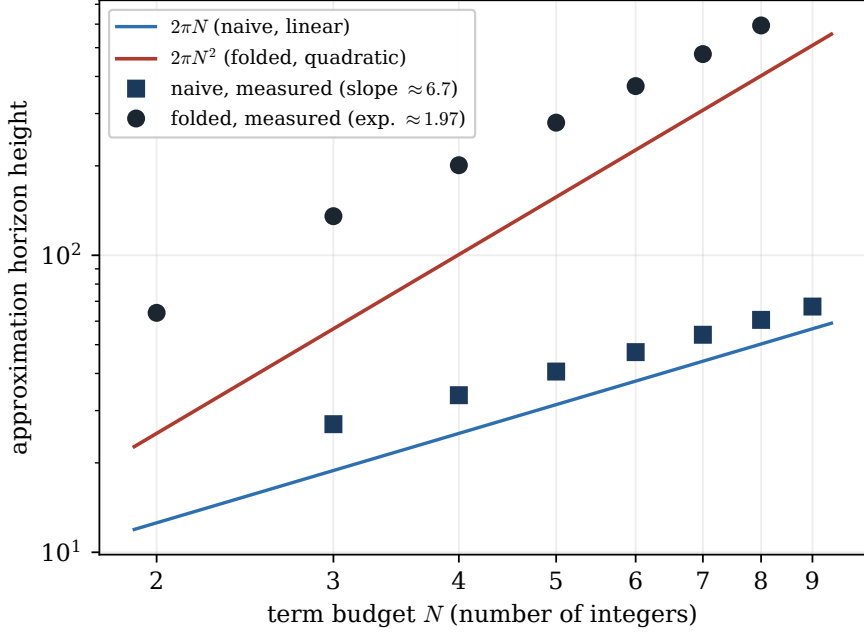


Figure 1: The approximation horizon (Level 1) versus term budget N , on logarithmic axes. Measured folded horizons (circles) follow the quadratic reference $2\pi N^2$ (fitted exponent 1.97); measured literal horizons (squares) follow the linear reference $2\pi N$ (fitted slope 6.70). The plotted heights are where a smoothed approximation error first crosses a fixed threshold; they characterise the approximation horizon, not the reconstruction count of Table 1. The small-budget offset of the folded points above the reference is a threshold effect: the data fix the exponent, while the prefactor is structural.

from the reconstruction results above; we include it only to situate the $\sqrt{\cdot}$ normalisation in a finite setting.

Proposition 7 (Gauss-sum amplitude). *Let p be an odd prime and χ a nontrivial multiplicative character modulo p , and let $g(\chi) = \sum_{x \bmod p} \chi(x) e^{2\pi i x/p}$. Then $|g(\chi)| = \sqrt{p}$.*

Proof. Classical [13, Ch. 8], [14]. Structurally, $\sqrt{p}$ is the modulus of the root number (ε -factor) in the functional equation of the Dirichlet L -function attached to χ [15], [16, Ch. 3]; the statement $|g(\chi)| = \sqrt{p}$ asserts that this root number is unimodular after the canonical $\sqrt{p}$ normalisation. $\square$

Figure 4 illustrates the amplitude: for $p = 99989$ all 99,987 nontrivial Gauss sums have modulus $\sqrt{p}$ to within 7×10^{-13} .

8 Scope, limitations, and outlook

What is and is not established. Level 1 is unconditional: the folded approximant has a quadratic approximation horizon $2\pi N^2$, sharp in mean square. Level 2 is conditional on simplicity of the critical-line zeros (Condition (A)) and on a local stability and separation condition on Z (Condition (B)); the latter can fail at close pairs, and we exhibited one. Level 3 is heuristic: the linear exponent is justified by zero counting (with constant $2\pi e$), while the sharp 2π is only observed. None of this constrains the real parts of the zeros or bears on the Riemann Hypothesis; the folded approximant reproduces zeros wherever they lie on the line.

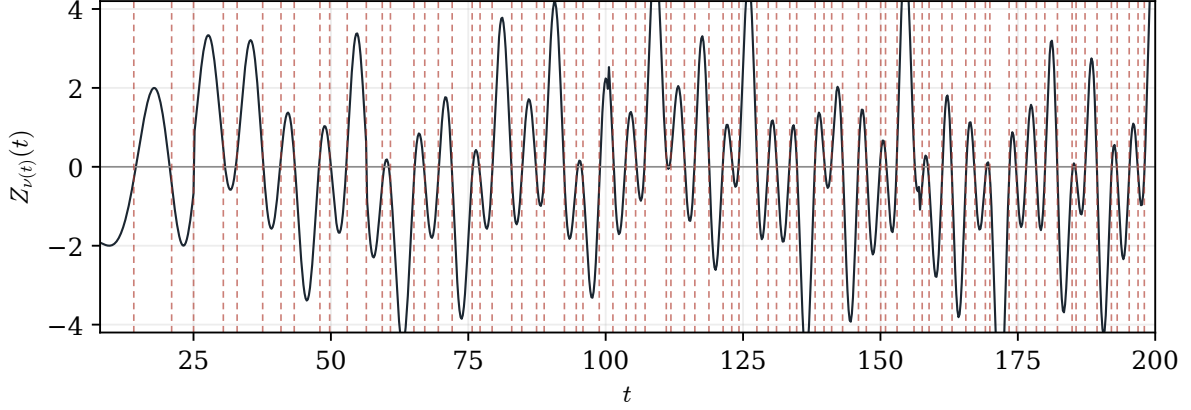


Figure 2: The folded approximant (2) at its prescribed term count $\nu(t) = \lfloor \sqrt{t/2\pi} \rfloor$ (growing from 1 to 5 over the window) vanishing at the ordinates γ (dashed), illustrating the reconstruction mechanism of Theorem 5 below the budget horizon.

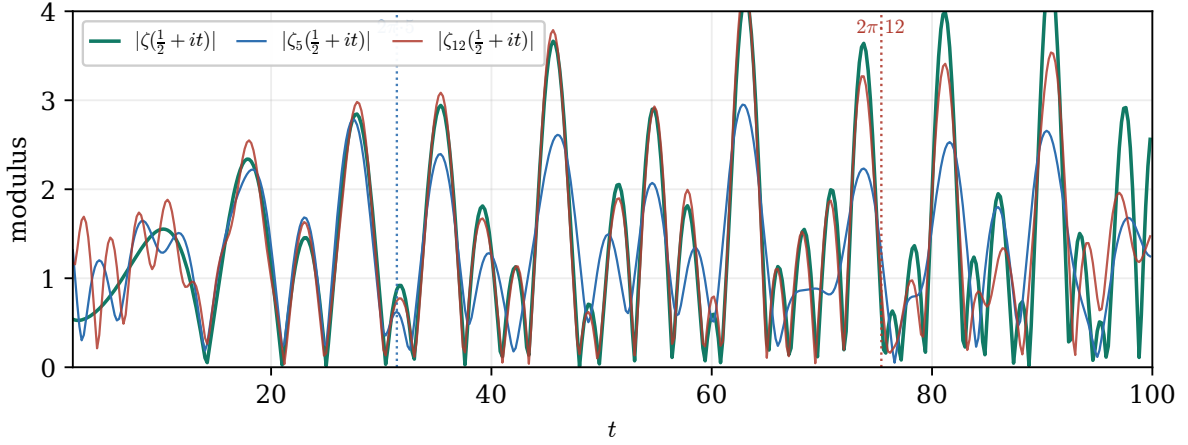


Figure 3: The literal truncation $|\zeta_N(\frac{1}{2} + it)|$ tracks $|\zeta(\frac{1}{2} + it)|$ only while $t \lesssim 2\pi N$ (dotted markers), then loses the zeros—the linear barrier of Heuristic 1.

Outlook. The natural way to strengthen Level 2 to an unconditional statement is to control the small values of Z between consecutive zeros—a question about the gaps and the density of close pairs. A separate direction is to ask whether the folded approximant can be realised so that its defining positivity is manifest on a finite-dimensional space, the setting of spectral positivity arguments [17, 18]; the present reach law fixes the analytic scale any such realisation must reproduce.

Acknowledgements

The author thanks colleagues in the Finite Ring Continuum discussion for sustained critical feedback, and acknowledges the use of open-source scientific software (`mpmath`, `NumPy`, `Matplotlib`).

Data and code availability

Scripts reproducing Table 1, the fitted exponents, and Figures 1–4 from scratch are provided as supplementary material.

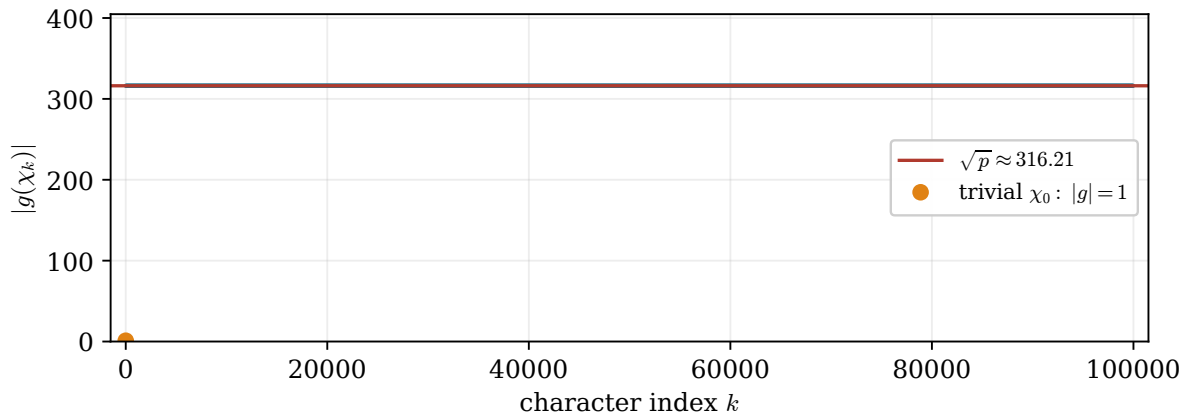


Figure 4: The finite analogue (Proposition 7): for $p = 99989$ the moduli $|g(\chi_k)|$ of all nontrivial Gauss sums equal $\sqrt{p} \approx 316.21$ to within 7×10^{-13} , while the trivial character gives $|g(\chi_0)| = 1$.

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